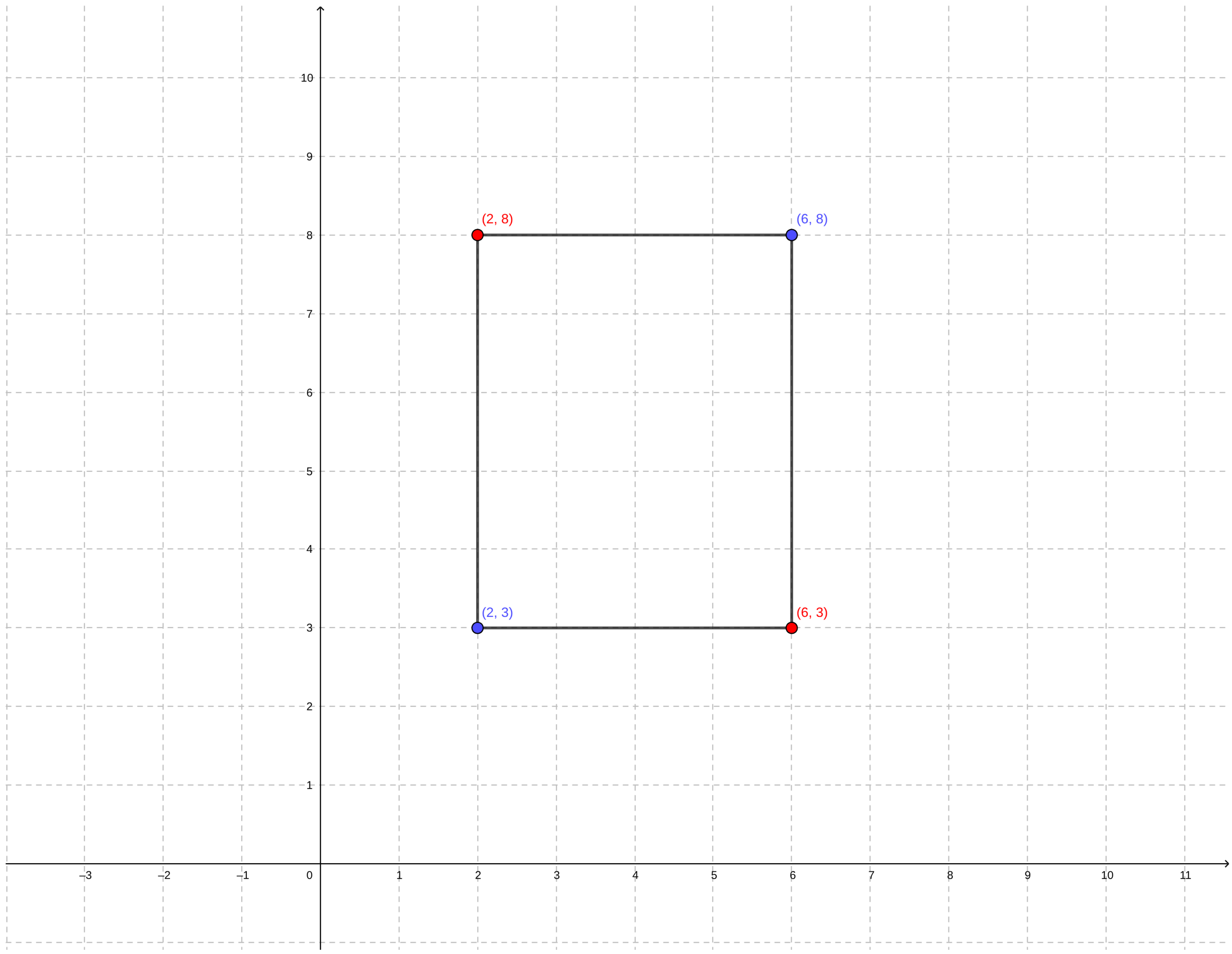
World and Object Representation

Exercises

1. **Bounding Box Definition**

- Given a 2D bounding box defined by $(x_{min}, y_{min}, x_{max}, y_{max}) = (2, 3, 6, 8)$, list all four corner coordinates.

  The bounding box corner will be given by $p_1 = (2, 3), p_2 = (2, 8), p_3 = (6, 8) and p_4 = (6, 3)$.



- Compute the area of the bounding box.

  The area of the bounding box is,

$$box_{area} = (x_{max} - x_{min}) \times (y_{max} - y_{min})$$

  Then,

$$box_{area} = (6 - 2) \times (8 - 3) = 20$$

  The $box_{area} = 20$.

2. **Bounding Boxes and Occupied Space**

- (a) Given a 3D bounding box with parameters $(x, y, z, l, w, h, \Psi) = (5, 3, 0, 4, 2, 2, 45^\circ)$, compute the volume occupied by the object.

  The volume box_{volume} occupied by the object is given by:

$$box_{volume} = l \times w \times h = 4 \times 2 \times 2 = 16$$

- (b) If the bounding box in (a) is rotated by $\Psi = 45^\circ$, sketch (or describe) how the occupied space differs compared to $\Psi = 0^\circ$.

  The heading Ψ is the measured angle of object local reference relative to the global origin coordinate $X - axis$, using the right hand rule. Which means that the rotation will be along the $Z - axis$. The Z rotation matrix is given by:

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

  Assuming the initial object direction is in $X - axis$, we can write,

$$P = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

  The rotation is,

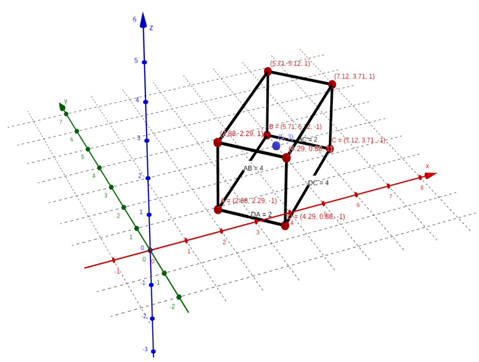
$$P_r = R_z(\theta) \times P = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

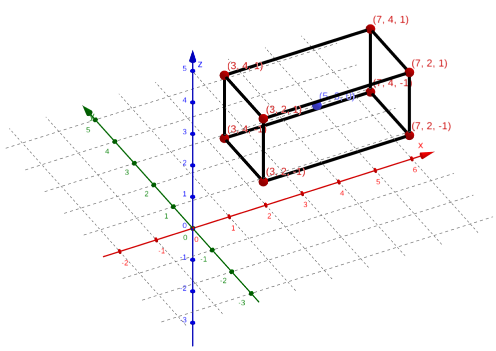
$$P_r = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \\ z \end{bmatrix}$$

  The translation matrix is,

$$T = \begin{bmatrix} 1 & 0 & 0 & x \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

  Applying the rotation and translation for each corner, $P' = T \times R_z \times P$,





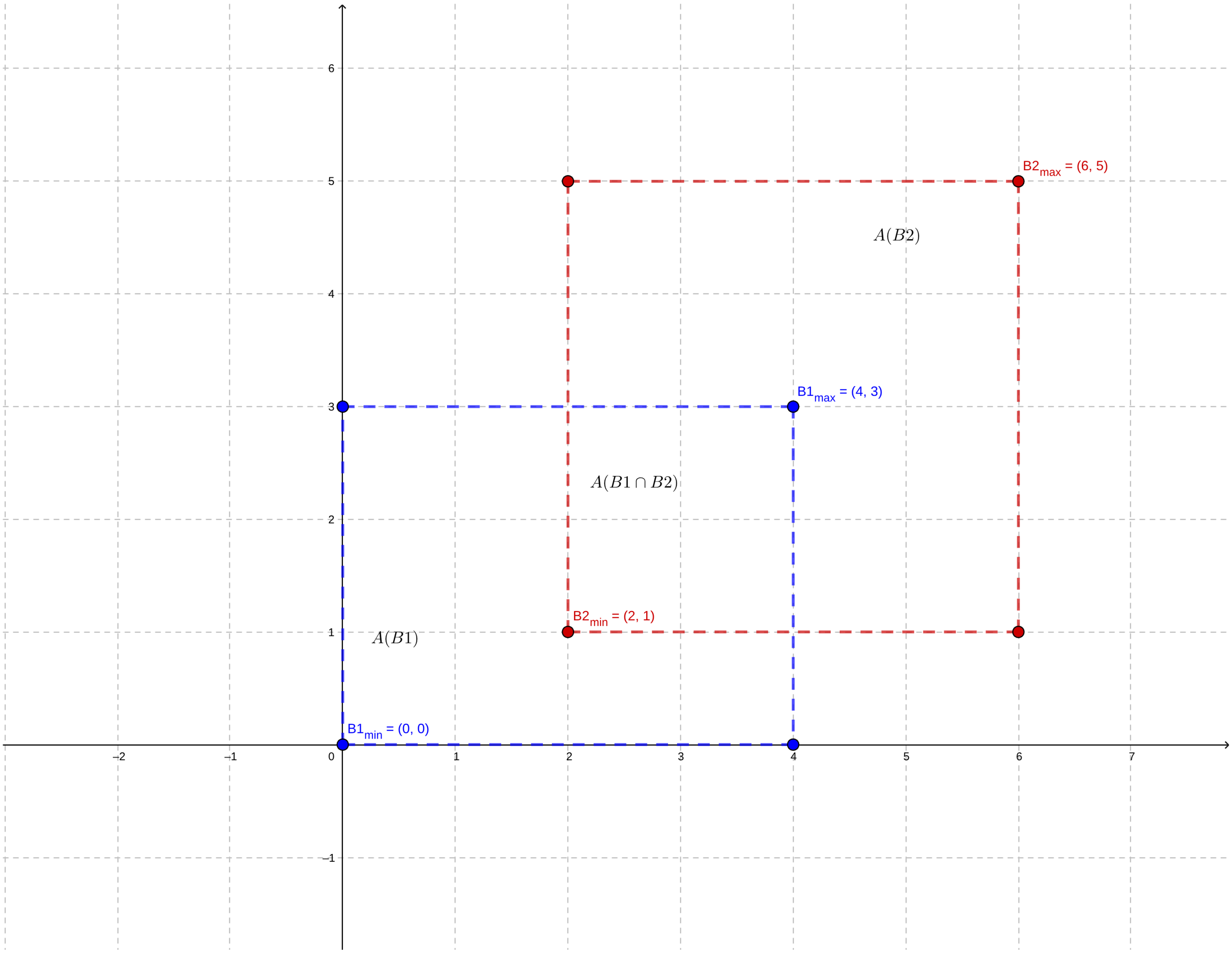
Show bounding box when $\Psi = 45^\circ$	*Show bounding box when $\Psi = 0^\circ$*

- (c) Two 2D bounding boxes $B_1 and B_2$ are defined as:

$$B_1 : (x_{min}, y_{min}, x_{max}, y_{max}) = (0, 0, 4, 3), B_2 : (2, 1, 6, 5).$$

Compute their intersection-over-union (IoU).

Given the two bounding boxes depicted in the image bellow, we will calculate the $A(B1 \cap B2)$ first,



The intersection are, denoted by $A(B1 \cap B2)$ can be calculated as,

$$A(B1 \cap B2) = (b1_{x_{max}} - b2_{x_{min}}) \times (b1_{y_{max}} - b2_{y_{min}})$$

$$A(B1 \cap B2) = (4 - 2) \times (3 - 1) = 4$$

The area for each bounding box is,

$$A(B1) = 4 \times 3 = 12,\ A(B2) = (5 - 1) \times (6 - 2) = 4 \times 4 = 16$$

The area of the union,

$$A(B1 \cup B2) = A(B1) + A(B2) - A(B1 \cap B2)$$
$$A(B1 \cup B2) = 12 + 16 - 4 = 24$$

Finally, we can calculate the IoU,

$$IoU = \frac{A(B1 \cap B2)}{A(B1 \cup B2)} = \frac{4}{24} = \frac{1}{6} \approx 0.1666$$

3. **Confidence and Uncertainty**

- A detector outputs a bicycle at 0.65 confidence. Discuss whether this uncertainty is more likely to be aleatoric or epistemic in the following scenarios:

  - (a) The bicycle is partially hidden behind a bus.

    This is more likely to be *aleatoric*. The bicycle is only partially observable by the detector.

  - (b) The detector was trained mostly on car and pedestrian classes, with few examples of bicycles.

    This is more likely to be *epistemic*. The detector need a better dataset with more classes for bicycles.

- A detector identifies a pedestrian at 0.85 confidence under foggy conditions. Which type of uncertainty (aleatoric or epistemic) is dominant, and why?

  The dominant uncertainty is *aleatoric*. The foggy condition is a environmental effect. Note that in such adverse condition, this high 0.85 confidence could be a false positive.

- An object is detected at 0.40 confidence. Suggest two possible ways to reduce epistemic uncertainty for this case.

  Increase the dataset for this specific scenario by collecting more data under similar condition. And create more scenarios, to increase detection confidence in new-but-similar scenarios where the detector will have good confidence even in cases it never saw, but was trained in a dataset with similar scenario.

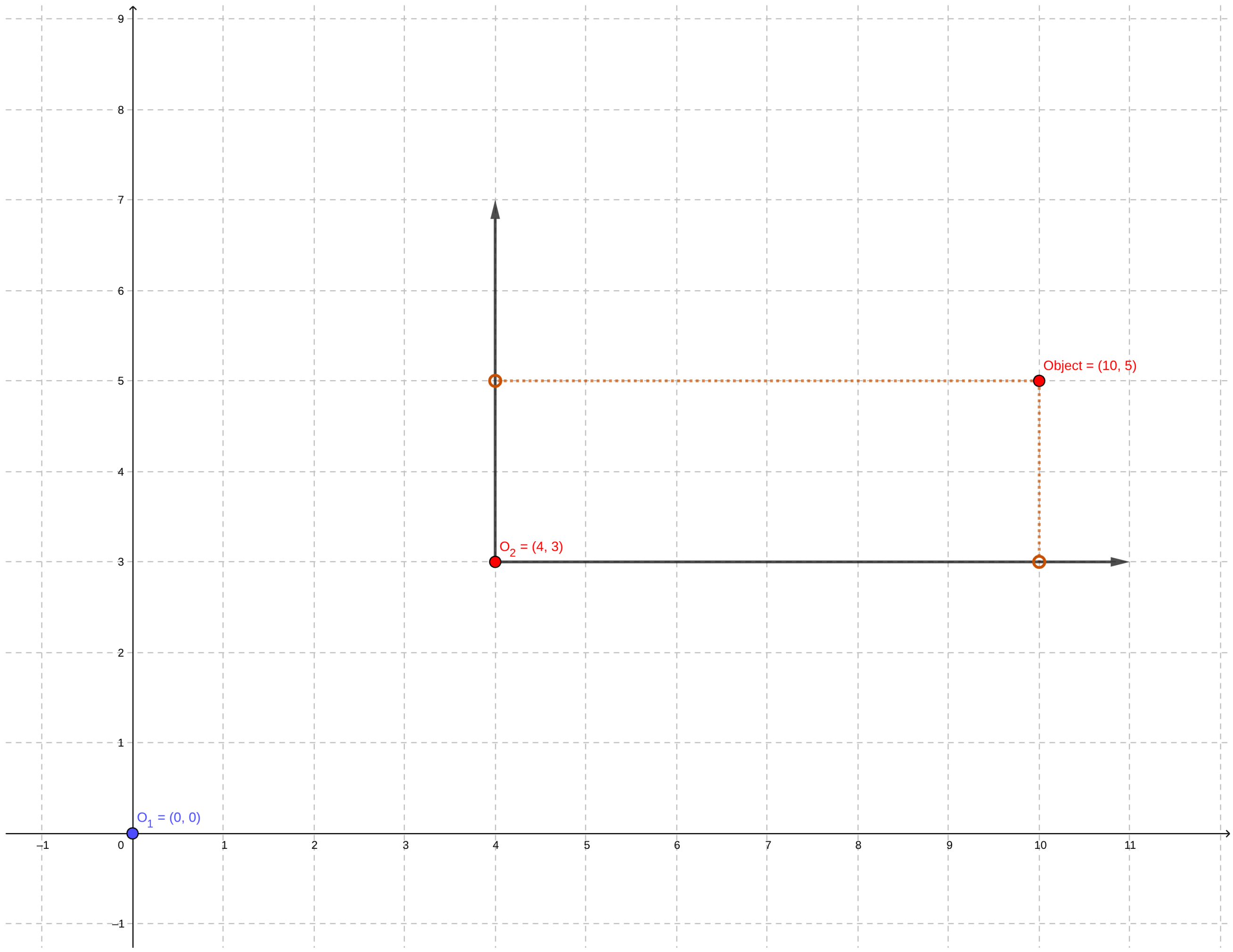
- Suppose the detector assigns high confidence (0.95) to a phantom detection caused by sensor reflection. Discuss why confidence alone may be misleading in safety-critical contexts.

  This is a false positive case where the detector sees something that is not there. Confidence is the indicator on how confident the detector is about its detection. Factors such as bad bounding box selection, lack of training data, faulty sensors, etc, can trick the detector into wrong classifications.

4. **Coordinate Transformation**

- An object is observed in the local frame of vehicle O_1 at position $(px, py) = (10, 5)$ m.
- The origin of O_2 is 4m east and 3m north of O_1.
- Compute the position of the object relative to O_2 assuming both vehicles use ENU coordinates.

  Given the $Object$ and origins O_1 and O_2, depicted bellow,



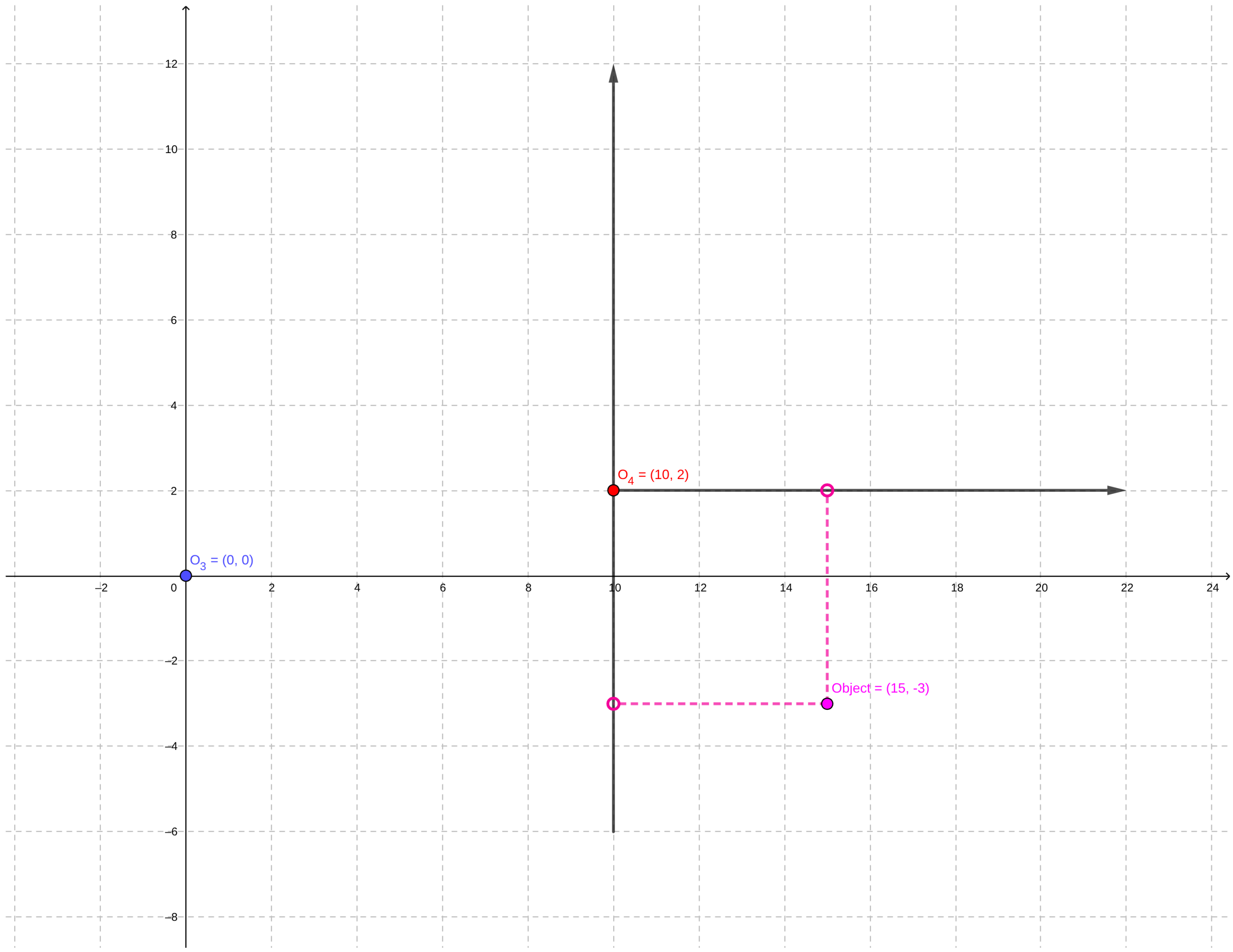
The position P' of the object relative to O_2 is,

$$P' = (x_{obj} - x_{O_2}, y_{obj} - y_{O_2}) = (10 - 4, 5 - 3)$$

$$P' = (6, 2)$$

- An observer O_3 detects and object at $(p_x, p_y) = (15, -3)$ m. Observer O_4 is located 10 m east and 2 m north of O_3. Express the object's coordinates in O_4's frame.

  Given the $Object$ and origins O_3 and O_4, depicted bellow,



The transformation matrix $T_{O_3 \to O_4}$ is,

$$T_{O_3 \to O_4} = \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

The position P' of the object relative to O_4' frame is,

$$P' = T_{O_3 \to O_4}(P) = T_{O_3 \to O_4} \times P$$

$$P' = \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 15 \\ -3 \\ 1 \end{bmatrix} = (5, -5)$$

Note: The position should be augmented by 1 row.

- Generalize the previous problem by deriving the transformation matrix $T_{O_3 \to O_4}$ for arbitrary displacements $(\Delta x, \Delta y)$ between observers.

  The transformation matrix $T_{O_3 \to O_4}$ can be expressed as,

  $$T'_{O_3 \to O_4} = \begin{bmatrix} 1 & 0 & (-10 + \Delta x) \\ 0 & 1 & (-2 + \Delta y) \\ 0 & 0 & 1 \end{bmatrix}$$

- If observer O_4 is additionally rotated by 30° with respect to O_3, update the transformation and compute the new object coordinates.

To better visualize the rotation, the can imagine the points rotating around a imaginary $Z-axis$. Given the 3D rotation transformation matrix for the $Z-axis$,

$$R_z(\theta) = R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The updated transformation to transform a object in O_3's coordinate to O_4 coordinate, must first rotate the O_4 then apply the transformation,

$$O'_4 = R(\theta) \times O_4 = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 10 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 10\cos\theta - 2\sin\theta \\ 10\sin\theta + 2\cos\theta \\ 1 \end{bmatrix}$$

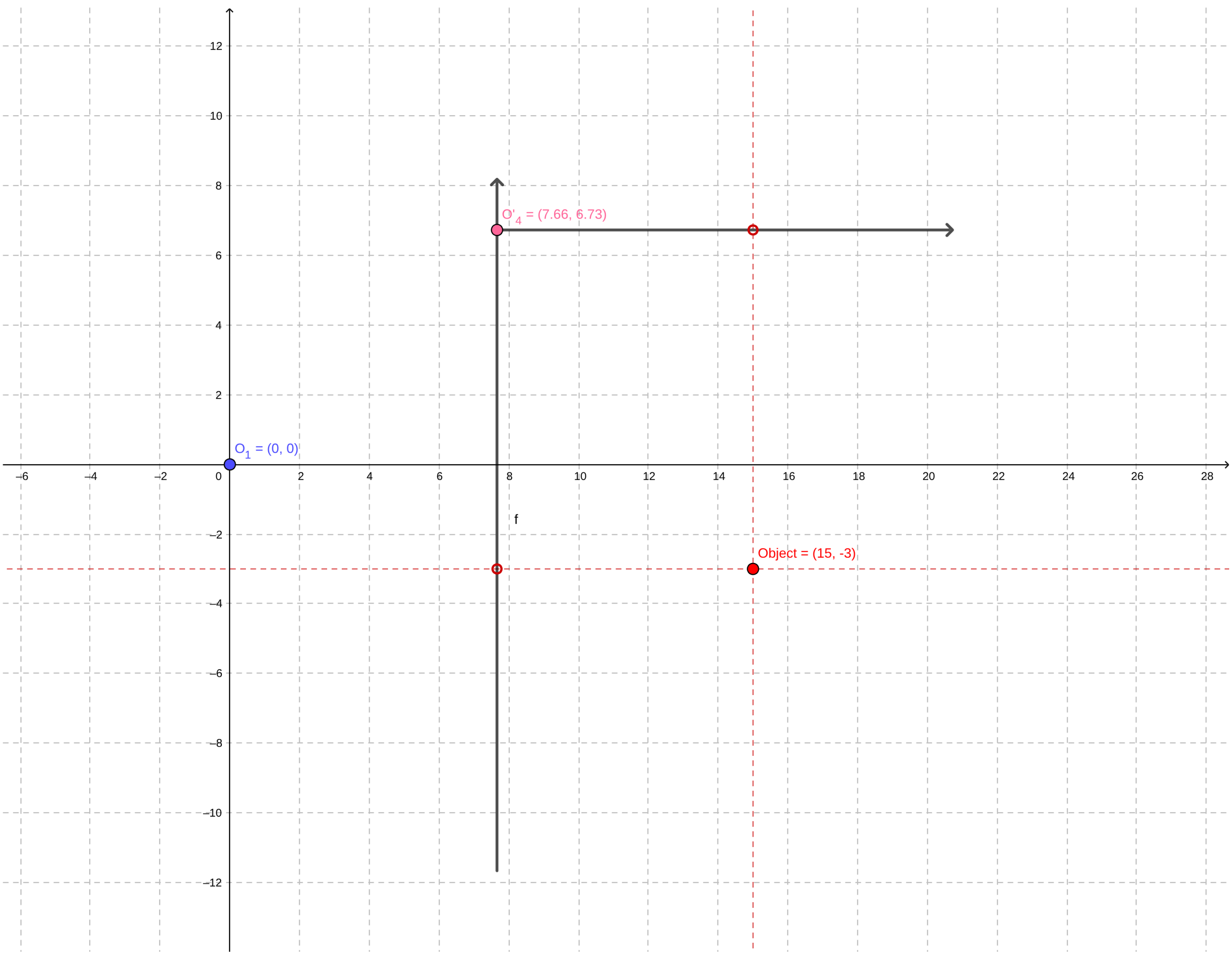
$$T_{O_3 \to O'_4} = \begin{bmatrix} 1 & 0 & -(10\cos\theta - 2\sin\theta) \\ 0 & 1 & -(10\sin\theta + 2\cos\theta) \\ 0 & 0 & 1 \end{bmatrix}$$

Applying the transformation $T_{O_3 \to O'_4}$ to $(px, py) = (15, -3)$ and making $\theta = 30^\circ$,

$$P'' = T_{O_3 \to O'_4}(15, -3) = \begin{bmatrix} 1 & 0 & -(10\cos 30^\circ - 2\sin 30^\circ) \\ 0 & 1 & -(10\sin 30^\circ + 2\cos 30^\circ) \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 15 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 15 - (10\cos 30^\circ - 2\sin 30^\circ) \\ -3 - (10\sin 30^\circ + 2\cos 30^\circ) \\ 1 \end{bmatrix}$$

Thus, the position of the object relative to O'_4, depicted on the image, is,

$$P'' \approx (7.34, -9.73)$$



5. **Motion Vector Propagation**

- A car modeled with the CTRV state vector:

$$x_k = \left[p_x, p_y, v, \Psi, \dot{\Psi}\right] = [0, 0, 20, 30^\circ, 0]$$

where speed is 20m/s, heading 30° and yaw rate 0.

- Compute is position after $\Delta t = 2$s.

Given a $\dot{\Psi} = 0$, the CTRV model fallback to a CV model, thus, we cal calculate the car position using,

$$x_{k+1} = x_k + v \cos \Psi \Delta t$$

$$y_{k+1} = y_k + v \sin \Psi \Delta t$$

Substituting the values into the equations,

$$P = \begin{bmatrix} 0 + 20 \cos 30^\circ \times 2 \\ 0 + 20 \sin 30^\circ \times 2 \end{bmatrix}^T = (34.64, 20)m$$

6. **Comparing Motion Models**

- Using the same initial conditions as Exercise 5, describe qualitatively how the predicted position would differ if the CTRA model were used with a longitudinal acceleration of $a = 2m/s^2$.

The CTRA model takes a motion vector $x_k = \left[p_x, p_y, v, \Psi, \dot{\Psi}, a\right]^T$, substituting the values, $x_k = [0, 0, 20, 30^\circ, 0, 2]^T$.

Same as the Exercise 5, there is a discontinuity when $\dot{\Psi} = 0$, thus, the CTRA model becomes,

$$x_{k+1} = x_k + \left[\left(v\Delta t + \frac{at^2}{2}\right)\cos\Psi\right]$$

$$y_{k+1} = y_k + \left[\left(v\Delta t + \frac{at^2}{2}\right)\sin\Psi\right]$$

Substituting the values,

$$P = \begin{bmatrix} 0 + \left[\left(20 \times 2 + \frac{2\times2^2}{2}\right)\cos 30^\circ\right] \\ 0 + \left[\left(20 \times 2 + \frac{2\times2^2}{2}\right)\sin 30^\circ\right] \end{bmatrix}^T = (38.01, 22)m$$

As the CTRV model does not take in consideration the longitudinal acceleration, the final position can be considerably different after some time.

- Likewise, compare with the CV model.

As noticed on Exercise 5, for this particular case where the $\dot{\Psi} = 0$, the observations made on the CTRA are the same for the CV. For a value of $\dot{\Psi} \neq 0$, the difference between CTRA and CV final positions would be much higher.

7. **Geopositioning and GNSS**

- A vehicle reports its position as $(lat, lon) = (52.5200^\circ N, 12.4050^\circ E)$ is WGS84. Explain how this can be approximated as a Cartesian coordinate in UTM.

The geodetic system can be represented onto a cartesian coordinate using the transverse mercator projection which "are formed by placing a cylinder in contact with a line of equal longitude also called a meridian (Figure 3) and projecting the earth's surface onto it."[2] The projection will have some distortions, however, it is minimized by the division of the earth's projection into 60 zones of 6° each.

- Two vehicles measure positions using GNSS, but both suffer from a 2m random error. Discuss the implications of this error when: (i) estimating absolute global positions, and (ii) estimating relative distance between the two vehicles when they are only 5m apart.

  For both cases, the GNSS adds some imprecision, due to conversions using geodetic systems. For (i), this error could lead to erroneous positioning that would impact a precise driving or maneuver, e.g. change road lanes, park, overtaking, etc. In (ii), if the vehicles rely only in the GNSS measurements, this could lead to accidents, e.g. vehicles stay in the same lane thinking they are 4m apart (assuming the error of both vehicles).

- A CAV network chooses to exchange relative Cartesian coordinates instead of geodetic coordinates. Justify this design in terms of uncertainty and computational cost.

  That are several reasons to choose Cartesian coordinate over GNSS:

  1. The GNSS has a slow rate of data transmission, that would be reasonable for long range distance traveling, where the position update rate can be slower, but for relative movement in close space requires a hight update rate;
  2. The GNSS coordinate system is not optimal for kinematic calculation, yielding slower computation as compared to cartesian coordinates.
  3. Continuously converting position between GNSS coordinate system and cartesian could lead to a build up of uncertainty.

8. **Time derivatives in rotation reference frames**

- Consider a scenario in which an autonomous vehicle has an IMU mounted at its center of rotation. Derive a differential equation that relates the measurements of the IMU, expressed in the vehicle's body frame, to the longitudinal acceleration of the vehicle.

Reference

[1] : https://geogebra.org
[2] : https://www.ccgalberta.com/ccgresources/report11/2009-410_converting_latlon_to_utm.pdf